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	Evaluations of risks involving alterations in the genome using whole animal systems are essential to missions in space. The lacZ transgenic mouse model is the only system available, to date, for the assessment of alterations in the genome in every tissue of the animal. In this model system, every cell of the animal contains multiple copies of an integrated but inert target transgene. Radiation-induced mutations can be measured and specific genetic alterations characterized using established protocols. Genetic alterations in tissues that are of high priority in NASA's Strategic Program Plans but are not accessible using conventional techniques, e.g., the central nervous system, can be evaluated using this model system. In addition, early (1-2 days) and late (up to 16 weeks) radiation responses in vivo can be examined in parallel by enumerating micronuclei in peripheral blood, evaluating chromosomal damage by using fluorescence in situ hybridization techniques, and induced gene expressions by using RT-PCR. Variation in genetic background has been shown to impact an individual's radiation sensitivities. The tumor suppressor p53 gene function has been shown to be radiation responsive and very important in the regulation of cell growth, proliferation, differentiation, and apoptotic		

Task Description:	signaling pathways in many tissues. LacZ transgenic mice with different p53 genotypes will be used to assess tissue-specific p53-dependent (or -independent) molecular and genetic mechanisms in radiation-induced damage resulting from exposure to particle beams in the energy range corresponding to space radiation. Shielding materials are being developed as countermeasures to protect astronauts against potential adverse radiation effects in long term manned space missions. In this current project, we aim to establish the use of this transgenic mutation model system in the testing of currently available shielding material to examine the efficacy of each substance in radiation protection. The research objective for our project is to use lacZ transgenic mice that are either Tp53 wild type, hemizygous or nullizygous to examine the impact of Tp53 genetic status on radiation sensitivity. The Specific Aims for our project include: (1) Characterize the dose- and time-dependent radiation-induced responses in lacZ transgenic mice after exposure to low-LET proton radiation. We hypothesize that the lacZ mutagenic potential of proton radiation is proton dose dependent and tissue specific. We also hypothesize that the initial level of micronucleated (MN) in peripheral blood reticulocytes (RET) and chromosome aberrations in circulating lymphocytes will increase as a function of dose. (2) Characterize the dose- and time-dependent radiation-induced responses in lacZ transgenic mice after 1 GeV iron particle irradiation. We hypothesize that lacZ mutant frequencies (MF) will increase as a function of iron ion dose for each tissue, but the level of induction of MF is dependent on the specific tissues analyzed. We also hypothesize that the initial level of MN in peripheral blood RET chromosome aberrations in circulating lymphocytes will increase as a function of dose For each of the above aims, we will examine the initial effects and long-term consequences of radiation exposure in tissues that are of high priority to NSBRI/NASA, namely the brain (CNS), spleen (highly proliferative tissue with stem cell populations), the hematopoietic system [immature polychromatic reticulocytes (RETs) and mature normochromatic erythrocytes (NCEs)] and lymphatic system. Using the same experimental animals, the transgene lacZ MF in brain and spleen, micronuclei in peripheral blood erythrocytes, and unstable chromosome aberrations in circulating lymphocytes, will be determined using conventional techniques. Comparisons of results from Aims 1 and 2 will provide new information on the LET-dependence of lacZ mutations in tissues, MN-RET in peripheral blood, and chromosome aberrations in lymphocytes at early and late times after irradiation. In addition, the impact of Tp53 genetic background on tissue-specific responses will be revealed. Knowledge gained in this ground-based work is directed at understanding tissue and organ response to space radiation and will aid both in the estimation of risk to astronaut health and the design of appropriate countermeasures to ameliorate the negative effects in space flights.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The space atmosphere is comprised of particle radiations in a wide spectrum of energies and charges, and radiation exposure poses a serious health hazard to humans in long-term manned space explorations. We are using the plasmid-based lacZ transgenic mouse mutation model system to examine the acute and long-term tissue-specific mutagenic responses of CNS and rapidly renewing organ systems after exposure to protons and HZE particles. We have cross-bred the lacZ animals with p53 knock-out animals in order to examine the impact of p53 genetic backgrounds on radiation responses. Transgenic mouse mutation model systems are accepted by regulatory agencies for toxicology studies. In addition to the aforementioned goals, we also aim to develop this mutation model system to be used to examine the effectiveness of targeted countermeasures, such as shielding materials, known radioprotective pharmaceutical or cytokines in the reduction of tissue-specific mutation frequencies or genetic damage in vivo. Analysis of genetic damage in different tissues in lacZ/p53 transgenics will allow us to test the efficacy of each countermeasure in the context of protection by reducing radiation damage and examine the impact of p53-dependent apoptosis in radiation protection. The results of this ground-based research project will provide information to characterize differences in tissue-specific mutagenic and clastogenic effects in vivo after particle radiation with different LETs, the acute and long-term consequences in tissue such as the brain, spleen and hematopoietic system and the effectiveness of available countermeasures in mitigating the effects of radiation exposure in a variety of tissues in vivo.
Task Progress:	Since the last progress report, we have made significant progress in examining the impact of Tp53 genetic background on radiation sensitivity in vivo. In our experiments, lacZ transgenic mice that are either Tp53 wild type (Tp53+/+), hemizygous (Tp53+/-) or nullizygous (Tp53-/-) were whole body exposed to either protons, X-rays or 1 GeV/n Iron ions. We assessed the temporal regulation of cytogenetic damage to the hematopoietic system by measuring the level of micronucleated erythrocytes (%MN-RET) in the circulating red cells or stem cell population in the bone marrow. Using tissue samples derived from the same animals, we also measured the effect of radiation damage at the genome level by quantitating the mutations (MF) in the genomically integrated transgene in two tissues with contrasting proliferative capacities, namely the brain and the spleen. Our multi-prong approach has provided insight into the dynamic processing of space-relevant radiation effects in the whole animal. Tp53 (-/-) and (+/-) animals, bred in-house at SRI International, were transported to Brookhaven National Laboratory (BNL) and exposed to graded doses of 1 GeV/n Iron ions. The %MN-RET was measured in the peripheral blood in a time course after HZE radiation exposure. The earliest cells to appear in the peripheral blood are those that were in the G2/M phase in the bone marrow at the time of radiation exposure. We have previously reported that the %MN-RET varies linearly as a function of dose and peaks at 48h after radiation exposure in the Tp53 (+/+) animals. New results show that %MN-RET in the peripheral blood shortly after radiation is dependent on Tp53 gene dosage. Animals lacking Tp53 gene function showed a delayed MN-RET output, independent of dose. At 48h after radiation, the level of %MN-RET appeared to vary linearly as a function of dose, saturating at doses >1 Gy, and was independent of Tp53 gene function. In addition to quantitating short term cytogenetic damage and processing by assessing the frequency of micronuclei in the peripheral blood, residual radiation effects in the bone marrow was evaluated by measuring MN and chromosomal aberrations (CA) in the bone marrow. A variety of aberrations were observed including gaps, breaks, acentric fragments, dicentrics, centric rings, robertsonian translocations. In Tp53(+/+) animals, the frequency of CA were within the control range at lower dose levels (0.1 and 0.5 Gy) for all time points, whereas at the highest dose level (2 Gy) there was a 3-fold increase in CA at 1 and 8 wks. The values returned to control levels at 16 wks after irradiation. In Tp53(+/-), there was a dose-dependent increase in CA at 1 wk. After 8 wks, the CA frequencies were within the control range at all dose levels. In Tp53(-/-) animals, there was no increase in CA at all dose levels compared to controls at 1 and 8 wks. These results suggest that Tp53 genetic background plays an important role in the determination of radiosensitivity of the stem cell populations and that most CA are readily removed during the recovery process in vivo. Brain and spleen tissues were collected at 8 wks post irradiation for mutant analysis. We have previously reported dose dependent increases in lacZ MF in the spleen tissues in Tp53(+/+) animals in response to Iron ions. Evaluation of lacZ MF in the brain from the same animals during this current year showed that, in contrast to the dose-saturation effect at high doses seen in the spleen tissues, the MF in the brain appeared to increase linearly with dose. Results from spleen tissues harvested from the Tp53(-/-) animals showed that the lacZ MF was independent of the dose of Iron ions, suggesting that damaged cells from the tissues in these

	animals were likely to be eliminated from the population and therefore, do not contribute to the long-term mutagenic potential for this tissue. We examined the effect of shielding animals with various thickness of polyethylene from 1 GeV/n Iron ion exposure. Transgenic mice were shielded with either 10 or 15 cm of polyethylene and exposed to either 0.1 or 1 Gy Iron ions. Early cytogenetic damage was assessed by enumerating the level of circulating MN-RET shortly after exposure and lacZ MF was evaluated at a later time up to 8 wks post exposure. We observed no difference between %MN-RET in animals with or without shielding at both dose levels. On the other hand, the lacZ MF in the brain and spleen with either 10 or 15-cm of shielding protected animals from radiation induced mutations in the reporter transgene. Results from these studies highlight the importance of endpoints considerations in the evaluation of the efficacy of potential radiation protectors.
Bibliography Type:	Description: (Last Updated: 04/26/2023)
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Presentation	Chang, P.Y., Bakke, J., Brune, J., Orduna,J. Rupa D.S. and Lin, S. "Impact of Trp53 genetic background, radiation quality and dosing on particle radiation-induced genetic damage in vivo" 3rd International Workshop for Space Radiation Research, Port Jefferson. Apr-2004
Presentation	Chang, P.Y., Bakke, J., Brune, J., Orduna,J. Rupa D.S. and Lin, S. "Impact of Trp53 genetic background on particle radiation-induced genetic damage in vivo, " Radiation Research Annual Society Meeting, St Louis, MI, Mar-2004
Presentation	Rupa, D.S., Rausch, L., Lin, S., Bakke, J., Orduna J., and Chang, P "Radiation-inducted chromosomal damage in transgenic mice with various p53 genotypes. " Environmental Mutagen Society, 35th Annual Meeting, Pittsburgh, PA Oct-2004